

Mercury's Tectonics Record a Damage Zone From the Caloris Basin-Forming Impact

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Key Points:

- We investigate if the Caloris basin on Mercury shows an impact damage zone
- Tectonic structures with Caloris basin-radial and -concentric orientations are identified across Mercury
- Several regions across Mercury, especially adjacent to the basin, are found to be consistent with a damage zone

Supporting Information:

Supporting Information may be found in the online version of this article.

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Citation:

Klimczak, C., Crane, K. T., Byrne, P. K., & Yazıcı, I. S. (2026). Mercury's tectonics record a damage zone from the Caloris basin-forming impact. *Journal of Geophysical Research: Planets*, 131, e2026JE009708. <https://doi.org/10.1029/2026JE009708>

Received 31 JAN 2026

Accepted 13 JUL 2026

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Abstract The Caloris basin is the largest of the well-preserved basins on Mercury and has left a dominant mark in the planet's geologic record. Previous analysis of orientations of a global compilation of Mercury's faults hypothesized that the Caloris basin-forming impact introduced a widespread damage zone from which basin-radial and -concentric fractures were reactivated as thrust faults by global contraction. Here, we test this hypothesis by expanding the analysis to straight rim segments of polygonal impact craters, which also show a global distribution across Mercury and, on the basis of crater degradation state, host useful stratigraphic information. We statistically assess orientations of fault segments and two populations of straight rim segments of polygonal impact craters: those with degradation characteristics consistent with pre-Caloris emplacement and the other with post-Caloris emplacement. We also extract basin-radial and -concentric orientations from the data sets that could host Caloris impact damage to investigate the spatial density of structures and locate geographic regions that stand out from observations reflecting randomly sampled orientations given the natural variation of the data. We find that comparison of statistically derived modes in the orientation distributions to basin-radial and -concentric orientations alone does not unequivocally reveal a damage zone, especially when combined in a global assessment. However, our analysis investigating regional details reveals that there may be a higher structure spatial density in latitudes and longitudes surrounding the basin and its antipode, and that a signature of the damage zone is likely asymmetric, hinting at an oblique impact from the northeast.

Plain Language Summary Meteorite impacts cause damage to the rock into which they impact. Impact damage is known from the study of laboratory experiments and field observations at impact craters on Earth to consist of many cracks that show patterns systematically oriented beneath and surrounding the impact site. Very large meteorite impacts (>1,000 km diameter) should have introduced widespread damage, but because no such impact structure is preserved on Earth, the nature of damage zones from the biggest of all impact events remains poorly characterized. We search for a damage zone around the 1,550-km-diameter Caloris basin, the largest of the well-preserved impact structures on Mercury. Because Mercury's tectonic record consists of tens of thousands of structures, some of them may have formed as part of a hypothesized Caloris basin damage zone. We study structures with arrangements around the Caloris basin that are consistent with those reported in damage zones and find that structure density and location show a signature of the damage. The damage zone is asymmetric and is preserved to the south and west of the basin. Taken together with previous geological observations, the impact forming the Caloris basin is likely to have been oblique, impacting from a northeasterly direction.

1. Introduction

1.1. Caloris Basin Geology

Mercury is a heavily cratered planet (Fassett et al., 2011; Herrick et al., 2018; Kinczyk et al., 2020) hosting upward of 80 large impact basins (Orgel et al., 2020). Among these impact basins is Caloris, which is considered the largest of the well-preserved basins on Mercury, with a diameter of ~1,550 km (Fassett et al., 2009, 2012; Orgel et al., 2020). The interior of the Caloris basin is covered by smooth plains up to 3.5 km thick (Ernst et al., 2015), attesting to volcanic modification of the basin floor (Murchie et al., 2008; Watters, Head, et al., 2009). These smooth plains were later tectonically modified. The basin center shows a basin-radial graben complex, referred to as Pantheon Fossae. Successively further away from the basin center, graben with basin-circumferential to polygonal patterns and basin-concentric shortening landforms are observed (Murchie

et al., 2008; Watters, Head, et al., 2009). Detailed mapping (Byrne et al., 2018; Cunje & Ghent, 2016) revealed that shortening landforms are also found closer to the center of the basin in both radial and concentric orientations. Watters, Head, et al. (2009) described the shortening landforms as cut by and thus older than the graben, but the basin-wide survey of cross-cutting relationships by Cunje and Ghent (2016) concluded that all structures formed as part of a continuous history instead of in discrete, separate stages.

Interestingly, Rothery et al. (2014) identified two separate clusters of volcanic vents near the southwestern rim of the basin that are aligned in a basin-radial orientation. This alignment was interpreted to have utilized deep-seated basin-radial faults, indicative of the deep structural inheritance of geological phenomena caused by the impact. Furthermore, long-term topographic adjustment of the basin (e.g., Watters et al., 2005) in conjunction with loading caused by the volcanic infill (Kennedy et al., 2008) has been invoked to explain structures with radial and concentric orientations within the basin. Certain loading scenarios also predict faulting to occur well beyond the basin interior (Kennedy et al., 2008).

Based on the unit descriptions by Guest and O'Donnell (1977) and Trask and Guest (1975) on Mariner 10 images of the eastern part of the basin, McCauley et al. (1981) defined a series of units exterior to the Caloris basin as the Caloris Group. With increasing distance from the basin, they include the Caloris Montes, and the Nervo, Odin, and Van Eyk Formations. All but the Nervo Formation were confirmed around the entire basin in the MERCURY Surface, Space Environment, Geochemistry, and Ranging (MERCURY Surface, Space ENVIRONMENT, GEOCHEMISTRY, and RANGING (MESSENGER) data (Fassett et al., 2009). The Caloris Montes are interpreted as the basin rim, consisting of numerous discrete blocks that are tens to hundreds of kilometers across. This unit comprises nearly continuous ranges with a clearly defined rim scarp or discontinuous smaller blocks and knobs. Based on the mapped location of the Caloris Montes from MESSENGER fly-by image data, the best-fit basin shape was determined to be elliptical with the long axis oriented in the ENE–WSW direction (Fassett et al., 2009). Although not an indicator of a highly oblique impact, the elliptical shape was interpreted as the impactor traveling in the direction of the long axis. Stereotopography of the Caloris basin (Oberst et al., 2010) combined with MESSENGER orbital image data highlighted the extent and morphology of the Caloris Montes, enabling the full extent of the basin to be mapped (e.g., Byrne et al., 2018; Byrne & Klimczak, 2025).

The Odin Formation was generally described as a hummocky plain unit abounding with knobs (Fassett et al., 2009; McCauley et al., 1981). Although earlier studies mapped the Odin Formation as an expansive unit surrounding the basin, more recent geological maps distinguish the basin exterior smooth plains units that are interpreted to be of volcanic origin, and exposures of Odin Formation interpreted as ejecta materials from the Caloris basin-forming impact that were not entirely superposed by smooth plains units (e.g., Mancinelli et al., 2016). Rothery et al. (2017) found that, although spectrally distinct, there are no clear boundaries between the plains units interior and exterior to the basin, with visible flow indicators into and out of the basin, suggesting a temporal overlap of unit emplacement and multiple volcanic source regions. A spectral comparison by Caminiti et al. (2023) of interior and exterior plains attributed heterogeneities between these units to decreases in the degree and depth of partial melting. In addition, crustal magnetization signatures coincide in part with the northern expanses of the Odin Formation, showing that impact materials could contribute to crustal magnetization (Hood, 2015; Kalski et al., 2025).

The knobs of the Odin Formation were found to be related to the Caloris impact but were later embayed by younger units (Ackiss et al., 2015). Detailed geologic characterization of hundreds of knobs convincingly demonstrated that they must be ejecta blocks that were subsequently modified by long-lived mass wasting, driven by volatile-loss (Wright et al., 2020). Knobs are most numerous west of the basin, showing the highest density in the southwest, where some are found ~1,000 km beyond the rim (Wright et al., 2020).

The Van Eyk Formation, as originally described by (McCauley et al., 1981), consists of lineated ridges and valleys interpreted to be impact sculptures as well as secondary craters with basin-radial geometries. Fassett et al. (2009) noted that the impact sculpting and secondaries are even more widespread than originally reported, attesting to the global influence of the Caloris impact. McCauley et al. (1981) characterized superposition relationships of complex craters with different degradation stages, which tied the global cratering record to impacts occurring before and after the Caloris basin-forming event.

Assessments of degradation states based on crater morphology remain widely used to categorize craters into classes for stratigraphic purposes (Herrick et al., 2018; Kinczyk et al., 2020; Spudis & Guest, 1988). Crater classes

typically range from “heavily degraded” to “very fresh.” Mercury generally follows three (e.g., Herrick et al., 2018; Mancinelli et al., 2016) or five classes (Kinczyk et al., 2020; Spudis & Guest, 1988). The categorization into five classes assesses the “crispness” of several aspects of the impact crater morphology, and considers the presence and morphological state of crater rays, rim and rim terraces, crater floor–wall boundary, the crater floor, the ejecta blanket, secondary craters, central structures such as peaks or peak rings, and the number of superposed small craters (Kinczyk et al., 2020). Following this scheme, class 1 craters are the most degraded, with all subsequent being successively less degraded until class 5, comprising craters that show the least degradation. Complex craters superposed onto the Caloris group are found to belong to classes 4 and 5. Some class 3 craters were also found to be superposed onto the Caloris group (McCauley et al., 1981), but no class 1 or 2 craters have been observed superposing the Caloris group, leading to the interpretation that the Caloris basin formed after all class 1 and 2 and some class 3 craters were emplaced. The global classification of complex craters by Kinczyk et al. (2020) was later applied to a global map of polygonal impact craters (Yazıcı et al., 2024), which showed that the straight rims of impact craters could be divided into two populations, one that presumably formed before and one that presumably formed after the Caloris basin was emplaced.

1.2. Tectonic Structures on Mercury

Mercury is not only a heavily cratered planet but also hosts thousands of large-scale tectonic structures (Byrne et al., 2018). The major driver of tectonics on Mercury is widely accepted to be global contraction, a process caused by the long, sustained cooling of the planet's interior (Solomon, 1976, 1977). Global contraction was accommodated in Mercury's lithosphere by widespread thrust faulting, expressed at the surface by numerous scarps and ridges (e.g., Byrne et al., 2018; Solomon et al., 2008). As a stand-alone process, global contraction is predicted to produce random thrust fault orientations (Melosh & McKinnon, 1988). However, several studies confirm that shortening landforms (and likely their underlying thrust faults) show strongly preferred and systematic orientations across the planet (e.g., Byrne et al., 2014; Byrne et al., 2018; Klimczak et al., 2015; Klimczak, Crane, & Byrne, 2025; Watters et al., 2015), thus requiring one or more tectonic processes to overlap with global contraction (Klimczak, Crane, & Byrne, 2025). Among these overlapping processes, the most extensively discussed include tidal despinning (e.g., Dombard & Hauck, 2008; Klimczak et al., 2015; Melosh, 1977; Melosh & Dzurisin, 1978; Pechmann & Melosh, 1979), mantle convection (e.g., King, 2008; Michel et al., 2013; Tosi et al., 2025; Watters et al., 2021), and tectonics influenced by the formation of the Caloris basin, either causing planetary reorientation (Matsuyama & Nimmo, 2009) or far-field consequences throughout the globally contracting lithosphere (Fleitout & Thomas, 1982; Thomas & Masson, 1984).

Tectonic structures on Mercury are also found within, and are related to, impact features. Basin tectonics show progressively greater complexity with increasing basin size, and consist of various configurations of normal faults expressed as grabens and troughs as well as thrust faults (e.g., Byrne et al., 2018). Ghost craters are volcanically flooded craters and basins demarcated by a thrust fault-related landform formed atop the buried crater rim; some of these ghost craters host randomly oriented small troughs in the center that are interpreted as small grabens (Klimczak et al., 2012). Normal and thrust fault-related landforms of various orientations are also found in a few peak ring basins with volcanic infill inside the peak ring (Blair et al., 2013). The normal faulting in ghost craters and peak ring basins is likely from extension caused by cooling of the volcanic unit within the impact structure (Blair et al., 2013; Freed et al., 2012). Several larger impact basins contain a system of thrust faults following the basin rim and showing outward fold vergences (i.e., with faults dipping toward the basin center) (Fassett et al., 2012; Fegan et al., 2017). The interface of the basin floor and volcanic infill was interpreted in these instances to have acted as a mechanical boundary along which shortening strain from global contraction was concentrated (Fegan et al., 2017). The Rembrandt basin, Mercury's largest, well-preserved basin in the southern hemisphere and the second-largest basin on the planet (Watters, Head, et al., 2009), shows collocated basin-radial sets of normal and thrust fault-related landforms near its center followed farther outward by basin-concentric normal and thrust fault-related landforms (Byrne et al., 2018; Ferrari et al., 2015). This system of structures is named Borobudur Fossae. The Caloris basin tectonic structures show the most complex geometric configurations, as described in Section 1.1. Because of their similarities in orientation and cross-cutting relationships, the structures within the Rembrandt and Caloris basins may have the same underlying causes, but no single process is able to explain all tectonic observations in those basins (Byrne et al., 2018).

The landforms formed by normal and thrust faulting are clear indicators of tectonics, as they are known to record extension and contraction, respectively, and because they have been preserved in the geologic record and remain

visible. However, hidden tectonic fabrics are possible to detect using polygonal impact craters, or PICs (Beddingfield et al., 2016; Cheng, 2026; Öhman et al., 2006, 2010). As excavation and modification during the cratering process can readily utilize preexisting crustal structures, crater rims commonly contain long, straight rim segments that, for complex craters, parallel such structures and result in polygonal planform crater shapes (Öhman et al., 2008, 2010). Preexisting structures very likely include cracks or joints up to km-scale faults that do not have pronounced surface expression but that are common in any lithosphere. On Mercury, a wealth of such cracks was not only utilized by PICs (e.g., Beddingfield et al., 2024) but also introduced by the impact cratering process (Kenkmann et al., 2014), increasing the degree of fracturing in its lithosphere (Klimczak, 2015). A global characterization of the planform shapes of impact structures between diameters of 20 and 400 km identified nearly 29,000 straight impact crater-rim segments longer than 10 km that were interpreted as having formed by utilizing joints in the lithosphere (Yazıcı et al., 2024).

Several mapping efforts on regional and global scales have documented the distribution of tectonic structures across Mercury. Thrust faults at various levels of detail were mapped globally (Bernhardt et al., 2025; Byrne et al., 2014; Man et al., 2023; Watters, 2021). Structures in the Borealis Planitia were mapped in detail by Crane and Klimczak (2019). Structures within the Caloris basin were mapped by Cunje and Ghent (2016) and Byrne et al. (2018) and in the Rembrandt basin by Ferrari et al. (2015) and Byrne et al. (2018). Tectonic structures within peak ring basins were mapped as part of the modeling by Blair et al. (2013) and ghost craters were recorded by Klimczak et al. (2012). From a combination of these efforts (i.e., Byrne et al., 2018; Byrne et al., 2014; Crane & Klimczak, 2019; Klimczak et al., 2012), Klimczak, Crane, and Byrne (2025) compiled a comprehensive global tectonic map containing nearly 42,000 fault segments belonging to ~18,500 tectonic landforms. Adding these structures to the mapped straight crater-rim segments of Yazıcı et al. (2024) yields a global tectonic map of thrust faults, normal faults, and straight crater rims indicative of joints, representing the most comprehensive tectonic map of the planet to date. Both the Yazıcı et al. (2024) and Klimczak, Crane, and Byrne (2025) studies used nearly the same methodologies for data collection and processing, such that the structural data, containing ~71,000 structure segments, compare well to one another. They are plotted in Figure 1.

1.3. Impact Damage Zones

The impact process introduces a profound number of fractures of multiple but systematic orientations beneath and around the crater. Shock compression and subsequent unloading during impact lead to failure accommodating extension via faulting and jointing (e.g., Ahrens & Rubin, 1993; Polanskey & Ahrens, 1990), but brittle failure during the impact process may be more complex, especially for highly oblique impacts (Stickle & Schultz, 2014). Fracturing also occurs in all stages of the impact process (Kenkmann et al., 2014). In the early stages, the shock wave may trigger small fractures but establish a densely spaced fabric, whereas later, during crater excavation and modification, some fractures coalesce to form km-scale faults. Three families of structures within impact damage zones were documented, including radial, concentric, and spall fractures. Two distinct subsets of radial fractures were characterized as conical and spoke fractures (Ahrens & Rubin, 1993). Of those, the farthest propagating cracks were found to be radial fractures (Xia & Ahrens, 2001) and the impact damage was noted to decrease with distance from the crater (Ahrens & Rubin, 1993). Shock-induced damage beneath craters formed by oblique impact was demonstrated to be larger in the downrange direction in both cratering experiments (Dahl & Schultz, 2001) and numerical models (Pierazzo & Melosh, 2000). Interestingly, impact damage was measured to noticeably reduce the elastic moduli of the target material (Moser et al., 2013), attesting to the importance of studying impact damage zones and their relevance for the properties of planetary lithospheres.

Geological field observations of impact craters on Earth show various aspects of fracturing within, beneath, and surrounding the crater (Kenkmann et al., 2014). Seismic investigations at the 1.2-km-diameter Meteor crater in Arizona have detected fractures down to 1 km beneath the crater floor (Ackermann et al., 1975), and hundreds of exposed fractures show radial and concentric orientations around the crater (Kumar & Kring, 2008), where the impact is interpreted to have reorganized pre-existing structures into radial and concentric orientations but also to have introduced additional fractures. The El'gytgytn crater in Siberia shows an increased fault lineament density within 2.7 crater radii before structures are indistinguishable from the background tectonic grid (Gurov et al., 2007), where some of the structures were also interpreted to have been part of a preexisting fault system that was rejuvenated during impact. Elevated porosities identified in a gravity survey underneath and surrounding the Granby and Tvären impact structures were attributed to fracturing as part of a damage zone that extends beneath and beyond the crater rim (Henkel et al., 2010).

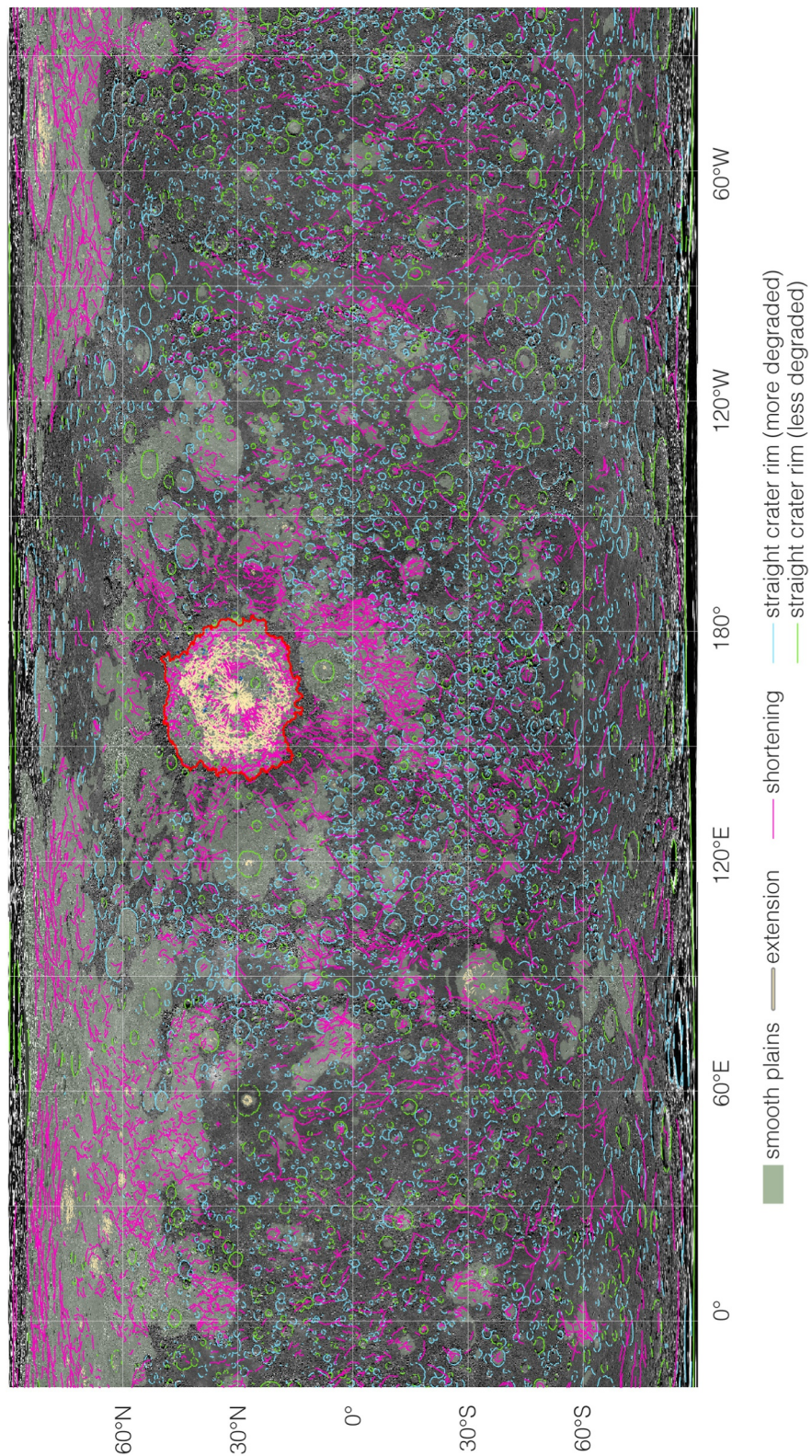


Figure 1.

Fractures at Lonar crater in India are also documented to occur in radial and concentric orientations surrounding the crater (Kumar, 2005). A seismic survey detected an asymmetric damage zone at the Lonar crater, which was attributed to an oblique impact with the projectile coming from a northeasterly direction (Kumari et al., 2024). Similarly, geologic lineament analysis using automated detection around the Manicouagan impact structure in Canada revealed radial and concentric structures surrounding the crater with a higher lineament density in the southwestern region around the crater (Fobert et al., 2025). This analysis suggested that the Manicouagan impact is likely a polygonal impact crater with the asymmetry in the lineament pattern arising from an oblique impact.

Tectonic observations around large impact basins on other planetary bodies have also led to interpretations of structures being impact related. On Mars, the area surrounding the 1500-km-diameter Argyre basin was found to be intensively fractured, as straight rim segments of polygonal impact craters indicate preferential radial and concentric orientations but also as conjugate shear fractures around the basins (Öhman et al., 2008). Searching for similar patterns of straight-rim segments of polygonal impact craters around the 450-km-diameter Rheasilvia basin on asteroid 4 Vesta showed some preferred orientations but without clear indications of basin-radial or -concentric structures (Cheng & Klimczak, 2025). However, structural mapping inside the Rheasilvia basin showed a disturbance in the otherwise systematic structural pattern in regions that coincide with those portions that superpose the Venenea basin, which was interpreted as a signature of damage inherited from the Venenea impact (Cheng & Klimczak, 2022).

On Mercury, Thomas and Masson (1984) described structural lineaments within but also surrounding the part of the Caloris basin that was imaged by Mariner 10. Lineaments within one radius surrounding the basin were found in basin-radial orientations. Although some of these lineaments were identified as being surficial, representing ejecta and secondary crater chains of the Van Eyk Formation, other structures were interpreted to be ejecta blocks bounded by rectilinear scarps. Based on their morphology and basin-radial orientations, these rectilinear scarps were interpreted to have been formed during or shortly after basin impact, with tectonic motions interpreted to have created the uplifted blocks, possibly also involving rejuvenation of preexisting structures. Numerical modeling efforts have explored several aspects of the Caloris basin-forming event (Potter & Head, 2017), basin structure (Gosselin et al., 2023), and post-impact evolution (Gosselin et al., 2025). The models in Gosselin et al. (2023) highlighted impact-induced failure and crustal block formation within and beneath the basin, but the size and nature of impact damage surrounding the Caloris basin remain to be fully characterized.

1.4. Hypothesis

Observations from hypervelocity impact experiments and field studies at terrestrial impact structures provide a set of characteristics of impact damage zones that serve as predictions for any impact damage zone on other planetary bodies. First, impacts introduce three families of fractures into the target rock surrounding the crater: those with orientations radial and circumferential orientations around the impact site and horizontal spall fractures (Ahrens & Rubin, 1993). Second, the damage decreases with distance from the impact site (Ahrens & Rubin, 1993), but fracturing should be noticeably increased compared to the background fracture density within one to perhaps more than two crater radii. Third, if the impact angle were high, the damage would be evenly distributed around the crater, whereas oblique impact angles produce asymmetric damage distribution around the crater, showing more damage in the downrange direction of the impact (Kenkmann et al., 2014).

No damage zone has ever been characterized for an impact structure as large as the Caloris basin. Notably, the orientations of faults around the Caloris basin were found to show preferential basin-radial, and -concentric orientations (Klimczak, Crane, & Byrne, 2025) (Figure 2), pointing to the existence of a large tectonic province related to the Caloris impact (Klimczak, Crane, & Byrne, 2025). It remains largely unknown if and how experimental results and observations from simple or complex craters scale up to the size of the Caloris impact. However, it is reasonable to hypothesize that impacts of that size not only cause damage but are likely to be

Figure 1. Global map of structural data on Mercury compiled from global shortening structures by Byrne et al. (2014), shortening structures in Borealis Planitia mapped by Crane and Klimczak (2019), structures within large impact structures and peak-ring basins (Byrne et al., 2018), and troughs within ghost craters (Klimczak et al., 2012), and straight rim segments of polygonal impact craters by Yazıcı et al. (2024). All shortening structures are shown in magenta, all structures accommodating extension are shown in beige, straight rim segments of heavily degraded impact craters are shown in light blue, and straight rim segments of moderately degraded to fresh impact craters are displayed in green. For reference, the Caloris basin is outlined in red and expanses of smooth plain units (Denevi et al., 2013) are shaded in olive green. The basemap is the MESSENGER monochrome morphology mosaic displayed in the Plate Carrée projection centered at 162.5°E longitude.

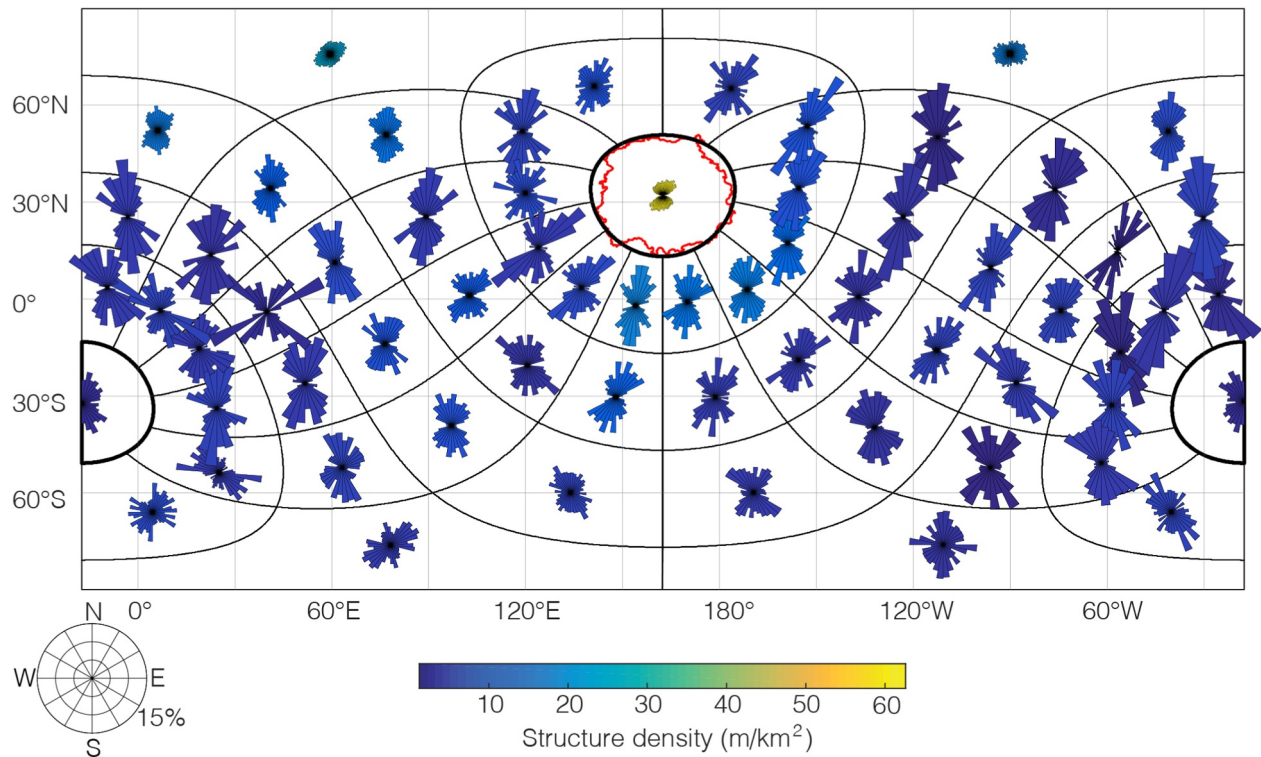


Figure 2. The distributions of fault orientations after Klimczak, Crane, and Byrne (2025) are shown as rose diagrams in arrangements around the Caloris basin in $30^\circ \times 30^\circ$ geographic regions separated by basin-radial and -concentric boundaries (black lines). The Caloris basin is outlined in red, and the geographic region of the basin and its antipode around which the geographic regions were drawn are bolded. Rose diagrams are color-coded from blue to yellow to represent structural areal density. All rose diagrams are scaled to one another. The outer perimeter of the scale, plotted at the bottom left, explains 15% of all observations. Note that rose diagrams have different bin degree-widths as determined by Silverman's Rule, which results in more bins and smaller diagrams with higher structure densities. The map is displayed in Plate Carrée projection, centered at 162.5°E longitude for comparison with Figure 1.

widespread. Following the predictions for impact damage zones listed above, we thus investigate the preservation and spatial extent of a damage zone from the Caloris basin-forming impact. To do so, we use the 71,000 structure segments from faults or rims of complex craters that we previously mapped across Mercury (Section 1.2, Figure 1). An impact damage zone would be recorded by a subset of these structures if fractures of the Caloris basin impact damage zone were either reactivated by tectonic processes, such as global contraction, to form thrust faults or utilized by the cratering process to form straight crater-rim segments.

2. Methods

In a geographic information system (GIS) using ArcGIS 10.8 and ArcGIS Pro, we combined the open-access global fault (Klimczak, Byrne, & Crane, 2025; Klimczak, Crane, & Byrne, 2025) and straight crater rims (Yazıcı et al., 2021, 2024) data sets (Figure 1). Both data sets contain information including fault and rim polyline azimuths, geodesic lengths, and, for straight crater-rim segments, the degradational class of the crater to which the segment belongs. For reference, we also plotted the Caloris basin outline (Byrne & Klimczak, 2025) and the global map of smooth plains (Denevi et al., 2013) alongside the crater rims and faults in Figure 1. To systematically extract data around the Caloris basin, we defined $30^\circ \times 30^\circ$ radial and concentric bins around the geometric mean center of the basin at 32°N and 162.5°E . Using the method described by Klimczak, Crane, and Byrne (2025), we constructed lines and circles as geodesic features with the advanced editing tools of ArcGIS Pro. Once all geodesic features were constructed, we used the “Feature To Polygon” function to convert the regions bounded by polylines to polygons, and assigned a unique identifier (i.e., a number) to each, leading to 60 geographic bins arranged around the Caloris basin with basin-radial and -concentric boundaries (e.g., Figure 2, black lines), as well as one bin for the basin itself and one for its antipode, which are outlined in bold in Figure 2. Basin-radial boundaries begin and terminate at the bins outlining the basin and antipodal region, whereas the basin-concentric bin boundaries are those that intersect with the radial bin boundaries. Next, we applied the

“Spatial Join” function to assign the unique identifier of the polygon to those fault and crater-rim segments whose center points were located within the polygon. Finally, we extracted the data containing structure type, length, coordinates of center point, azimuth, and geographic bin for each structural segment for our analysis.

We previously analyzed orientations of faults with geographic binning configured around the Caloris basin (Klimczak, Crane, & Byrne, 2025), and we applied the same analysis and visualization to the crater orientation data separated into moderately degraded to fresh straight crater-rim segments and heavily degraded straight crater-rim segments. Moderately degraded to fresh straight crater-rim segments are those that combine observations from crater classes 3 to 5 and are interpreted to have formed after the Caloris basin-forming event. Heavily degraded straight crater-rim segments combined with observations from crater classes 1 and 2 are interpreted to have been emplaced prior to the Caloris basin-forming event. First, we used a kernel density estimation (KDE) and applied the Multimode Package in *R* (Ameijeiras-Alonso et al., 2021) to calculate the quantity and values (i.e., orientations) of statistically significant modes for each bin of fault segments. We used the “Locmodes” function in *R* (Ameijeiras-Alonso et al., 2019) to determine the respective strength of each mode from the KDE. We also applied Silverman’s Rule of Thumb (Silverman, 1986) in *R* to calculate the optimal bandwidth from the size, standard deviation, and interquartile range for the distribution of azimuths in each geographic bin. We tabulated the modes for faults, moderately degraded to fresh straight crater-rim segments, and heavily degraded straight crater-rim segments in Tables S1–S3 of Supporting Information S1, respectively.

Again following the method described in detail in Klimczak, Crane, and Byrne (2025), we visualized the orientations of faults and crater rims using length-weighted rose diagrams with a degree-width of bins statistically determined to best highlight the variation in the data. We color-coded the rose diagrams based on structure density, as determined by the total length of structures within each geographic region divided by the area of the region. We plotted each rose diagram to scale to one another at the center of each geographic region for structural interpretation of orientations across Mercury (Figures 2 and 3).

We determined the full range of orientations of the polylines that bound the geographic regions. For each of the regions, we then determined to 10° accuracy the numeric ranges of orientations that qualify as concentric and radial with respect to Caloris. We compared the modes returned by our statistical analysis to the range of basin-radial and -concentric orientations (Tables S1–S3 in Supporting Information S1). In addition, we created a separate shapefile containing all faults and all moderately degraded to fresh straight crater-rim segments and added columns to the attribute tables to mark those segments that qualified as basin-radial and -concentric. To do so, the Structured Query Language (SQL) in ArcGIS Pro was used to search each geographic region with the range of orientations of interest. The “Calculate Field” function was then used to mark the returned structural segments as either radial or concentric. We also marked those structures within the basin and at the antipode. The shapefile containing the structure type and their relationship in orientation with respect to the Caloris basin is made available as a data set (Klimczak et al., 2026).

After all geographic regions were queried and all structures of basin-radial and -concentric orientations were identified, we computed data points at every 1 km along fault segments and moderately degraded them to fresh straight crater-rim segments to investigate the spatial density of structures of a potential damage zone. From these data points, we computed histograms with a 2° binning to display the frequency of latitudinal and longitudinal observations. The basin-concentric structures and their distribution of latitudinal and longitudinal observations are presented in Figure 4a. The same observations are plotted in Figure 4b for basin-radial structures.

Finally, we assessed whether the radial and concentric orientations are likely part of an impact damage zone. For each of the three populations of structural segments and for each geographic bin, we computed the percentage of observations that are basin-radial and -concentric. Although fault segments and moderately degraded to fresh straight crater-rim segments could contain fractures that originally formed as part of the Caloris impact damage zone, the heavily degraded straight crater-rim segments presumably formed prior to the Caloris basin-forming impact and therefore should not contain structures formed by the impact, providing us with a control distribution. We assumed that any basin-radial and -concentric structures in this population were randomly sampled and that their percentages reflect the value of arbitrarily extracted orientations, even in the population of fault segments and moderately degraded to fresh straight crater-rim segments. We thus compared the percentages of fault segments and moderately degraded to fresh straight crater-rim segments to the percentage of the heavily degraded

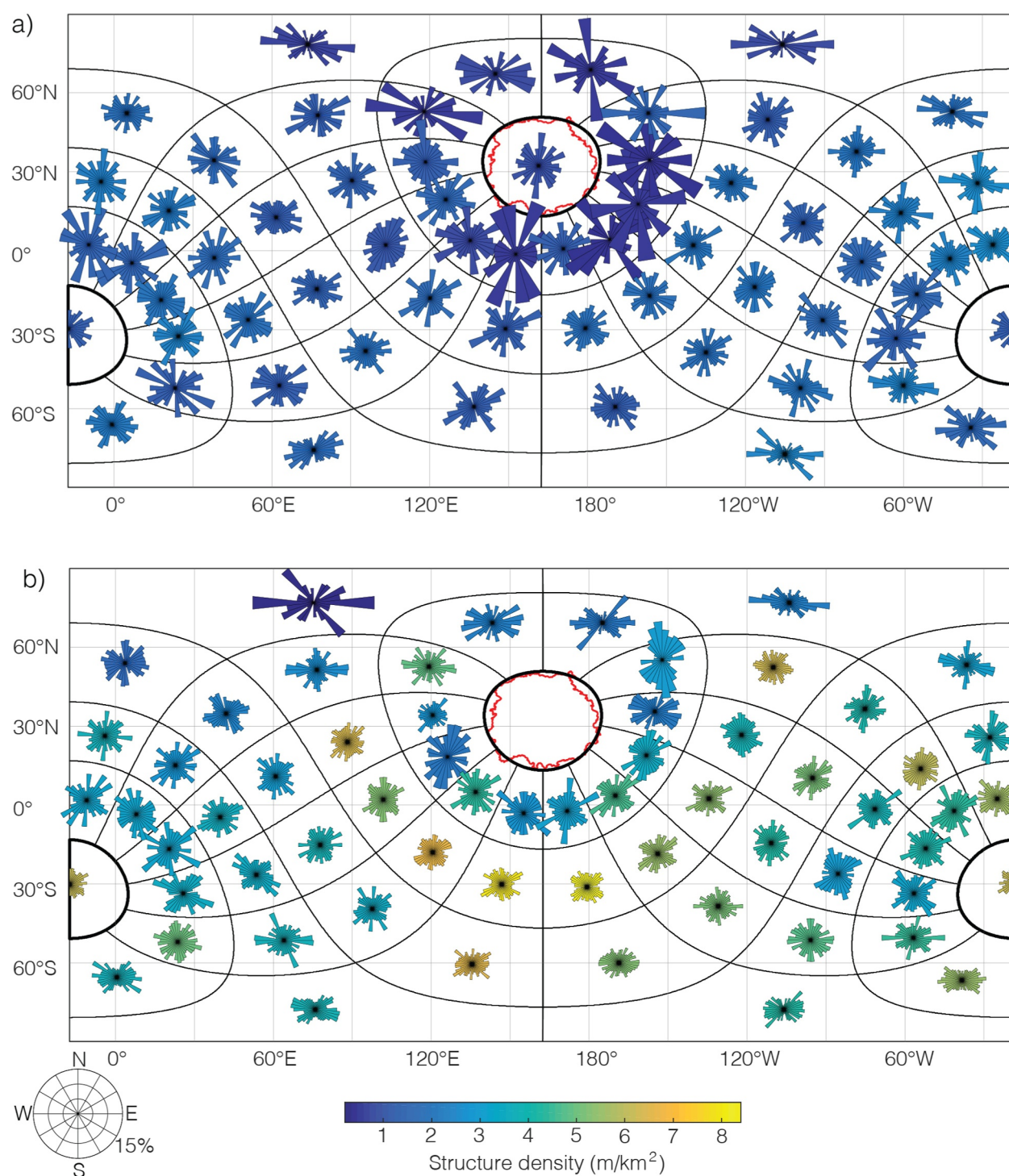


Figure 3. The distributions of straight crater rim orientations are shown as rose diagrams plotted in arrangements around the Caloris basin, using the same annotations as in Figure 2. The maps are also displayed in Plate Carrée projection, centered at 162.5°E longitude for comparison with all other figures. (a) Distributions of orientations of moderately degraded to fresh straight crater-rim segments. (b) Distributions of orientations of heavily degraded straight crater-rim segments.

straight crater-rim segments, marking only those geographic regions where the percentages of fault segments and moderately degraded to fresh straight crater-rim segments exceeded the percentage of heavily degraded straight crater-rim segments.

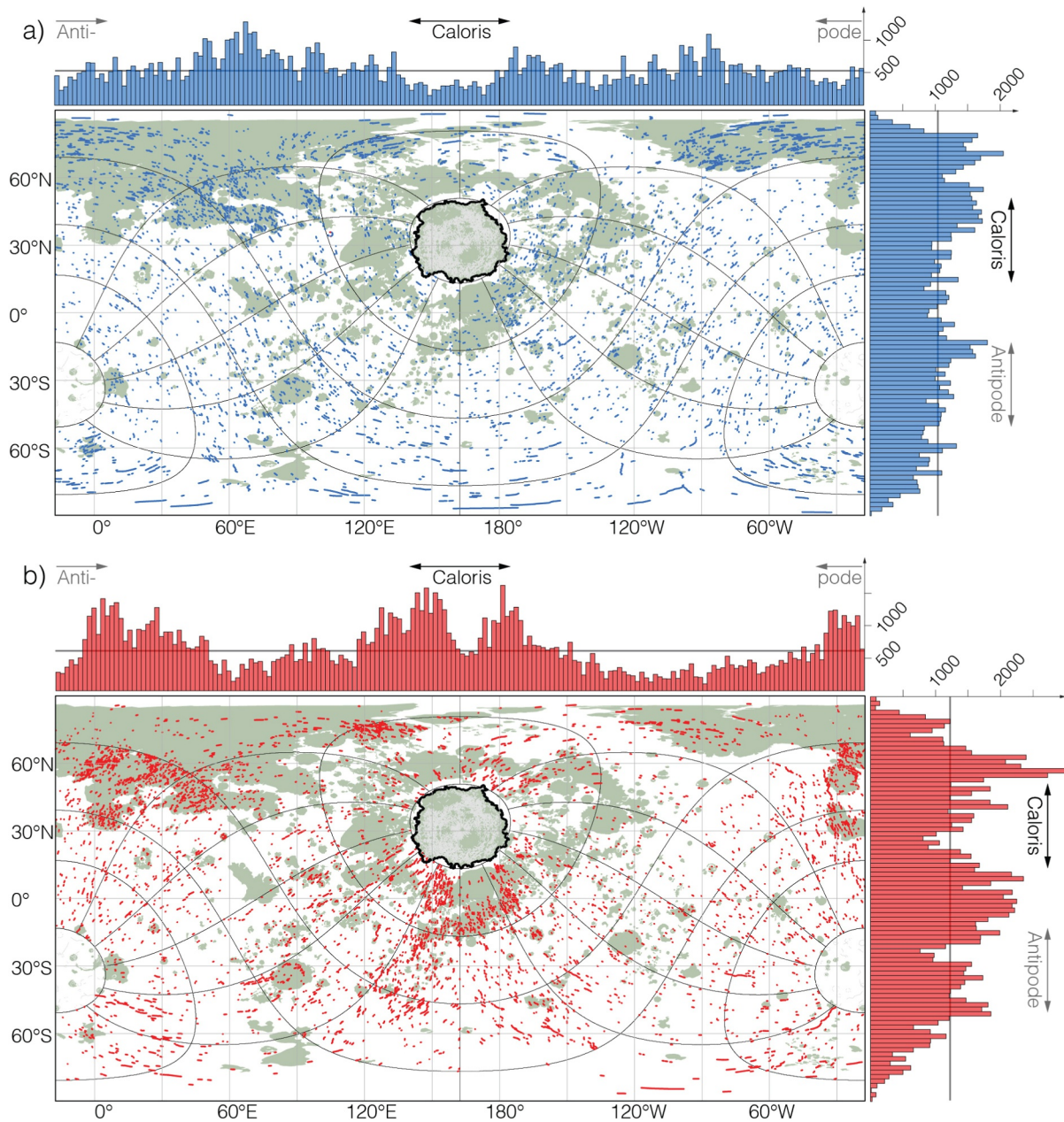


Figure 4. Global map of structural segments of faults and moderately degraded to fresh straight crater-rim segments only showing (a) concentric and (b) radial orientations with respect to the Caloris basin, outlined here in black. The maps are also displayed in Plate Carrée projection centered at 162.5°E longitude for comparison with all other figures. Histograms plotted along the longitudinal and latitudinal axes of the map show the frequency of (a) concentric and (b) radial structures with a 2° bin width.

In addition, we also computed the percentage of the range of orientations considered basin-radial and -concentric. Some geographic regions have a narrow range of basin-radial and -concentric orientations, for example, only 20° out of 180° (11%) were found to be radial and concentric, whereas other geographic regions showed rather wide ranges, for example, as much as 90° out of 180° (50%). Greater ranges of such orientations also resulted in the extraction of a larger percentage of segments from the data set, showing that within the data some structures are randomly consistent with the structural orientations of interest. We therefore interpreted the extracted radial and concentric orientations to only likely contain fractures related to the Caloris damage zone if their percentage exceeds that of the sampled range. In comparing these percentages, then, we only marked those geographic regions where the percentages of extracted fault segments and moderately degraded to fresh straight crater-rim

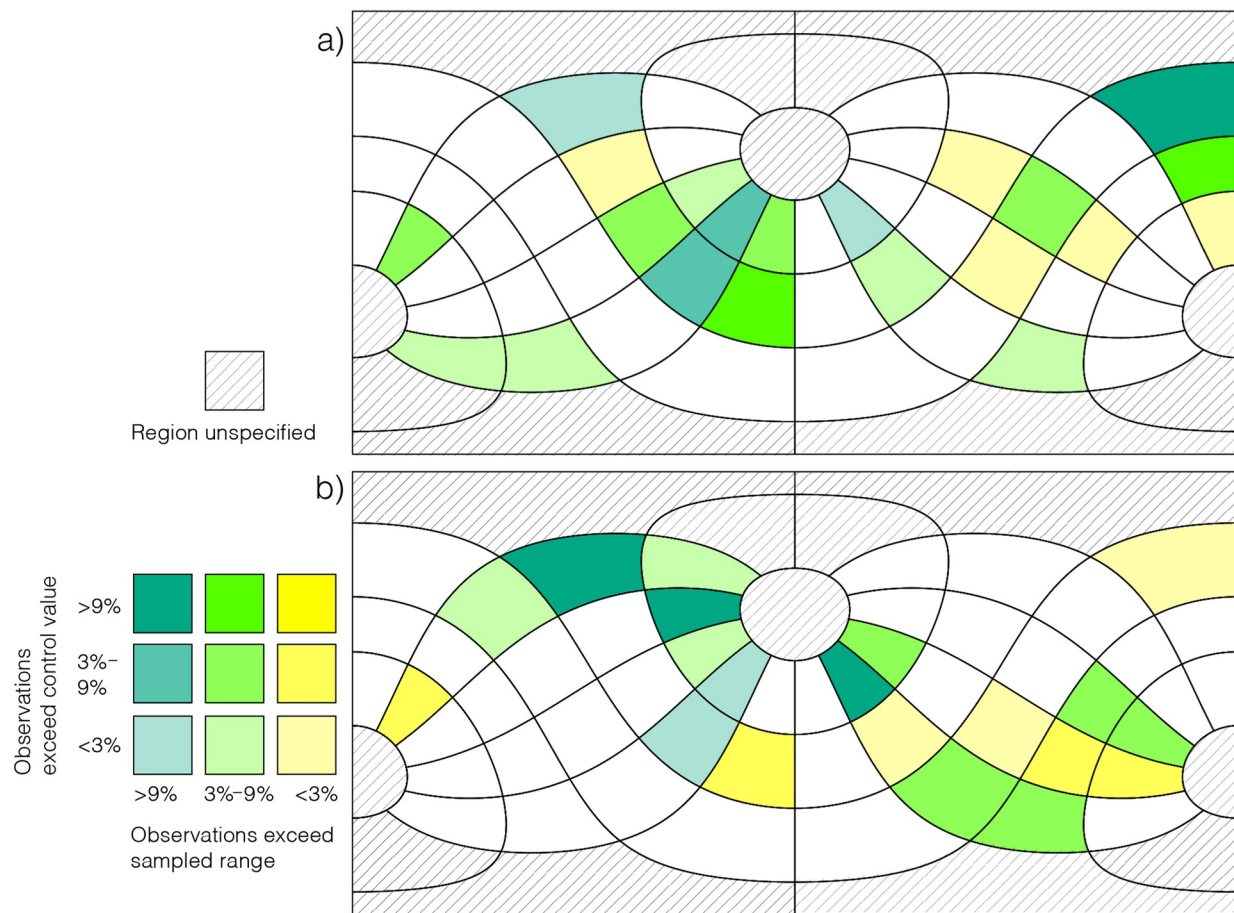


Figure 5. Bivariate maps of (a) fault and (b) moderately degraded to fresh straight crater-rim segments highlighting those geographic regions where Figures 1–4 can be interpreted to contain a signature of Caloris basin impact damage. Colored fields show regions where observations exceeded both the control value and the sampled range. Colors correspond to the percentages of basin-radial and -concentric structures in the (a) fault and (b) moderately degraded to fresh straight crater-rim segments that exceed that of the heavily degraded crater-rim segments. The color change from yellow to dark green indicates an increase in these percentages. The increasing color saturation reflects the percentages of observations of basin-radial and -concentric structures in the (a) fault and (b) moderately degraded to fresh straight crater-rim segments that exceed the percentage of the range that was identified as consistent with basin-radial and -concentric orientations within that bin.

segments exceeded the percentage of the sampled range. We considered the geographic bins to show a signature of the Caloris impact damage zone only if both the percentages of control value and sampled range were exceeded. The results of the comparison are plotted separately for fault segments and moderately degraded to fresh straight crater-rim segments as bivariate maps in Figure 5.

3. Results

We displayed the full set of orientations of the analyzed structural segments as rose diagrams within the 60 geographic regions arranged around the Caloris basin and for the basin itself and its antipode, showing faults (Figure 2) as well as moderately degraded to fresh straight crater-rim segments (Figure 3a) and heavily degraded straight crater-rim segments (Figure 3b). For all three data sets, we identified statistically significant modes in all geographic regions (Tables S1–S3 in Supporting Information S1). For the fault data, it was previously noted by Klimczak, Crane, and Byrne (2025) that, although several strongly preferred orientations are contained within the data, many rose diagrams show either radial, concentric, or bimodal distributions of radial and concentric preferred fault orientations around the Caloris basin (Figure 2; Figure S1 in Supporting Information S1), with such combined orientations being most noticeable in the hemisphere in which the basin formed. The most notable radial orientations occur just outside the basin to the southwest, where three geographic regions have the strongest modes and >40% of the data coincide with radial orientations (regions 5–7, Figure S1 in Supporting Information S1). The antipodal hemisphere also shows some orientations consistent with basin-radial and -concentric

orientations. These observations were previously interpreted to indicate that the Caloris basin-forming impact produced an extensive damage zone in which fractures were subsequently reactivated by faulting (Klimczak, Crane, & Byrne, 2025), motivating this study.

For the moderately degraded to fresh straight crater-rim segments (Figure 3a), bimodal and polymodal orientations are identified across all geographic regions, showing at least two but up to four statistically significant modes. Similar to our observations for the fault segments, these bi- and polymodal orientations likely attest to multiple superposed tectonic patterns that exist on Mercury (Klimczak, Crane, & Byrne, 2025). In the majority of the geographic regions, there are modes that show either radial, concentric, or radial and concentric preferred fault orientations around the Caloris basin (Table S2 in Supporting Information S1), but these orientations are difficult to discern in the pattern displayed by the rose diagrams (Figure 3a) as other dominant structural trends are also contained in the data. Many regions, especially directly next to Caloris, show the strongest modes as radial or concentric to the basin (e.g., regions 3–5, 9, and 12 in Figure S2 of Supporting Information S1). In fact, all but one of the regions directly adjacent to the Caloris basin contain radial modes, with the one region (region 7) showing its strongest mode just outside the range consistent with basin-radial orientations.

For the heavily degraded straight crater-rim segments (Figure 3b), we also detected bimodal and polymodal orientations across all geographic regions, showing at least two but up to four statistically significant modes. This finding indicates that multiple tectonic patterns are contained in all of the analyzed structural data. As with fault and moderately degraded to fresh straight crater-rim segments, heavily degraded straight crater-rim segments also show radial, concentric, or radial and concentric preferred fault orientations in many regions around the Caloris basin (Table S3 in Supporting Information S1). Whereas the basin-radial and -concentric orientations are not represented as much by the strongest modes in this data set, and somewhat fewer basin-radial modes are confirmed in the geographic regions directly bordering the Caloris basin, there is no clear difference apparent in orientation trends when compared to the fault and moderately degraded to fresh straight crater-rim segments. This finding shows that some proportion of radial and concentric orientations are coincidentally consistent with radial and concentric orientations, and that interpretations of orientations based on statistically determined modes alone are not sufficient for characterizing the presence of a Caloris damage zone.

We extracted all structural segments that matched the range of radial and concentric orientations in each geographic region and for all three data sets. We displayed the concentric and radial orientations of the combined data for fault and moderately degraded to fresh straight crater-rim segments in Figures 4a and 4b, respectively. We consider these structures to be candidate structures of the Caloris basin damage zone. We use the extracted data for the heavily degraded straight crater-rim segments for further analysis, but do not display them, as they are presumably older than the Caloris basin and thus unlikely to have primarily formed from the Caloris impact. The number of extracted structures with basin-radial and -concentric orientations constitutes global averages of 35.5% and 37.5% of all segments of faults and is moderately degraded to fresh straight crater rims, respectively. The global average for basin-radial and -concentric orientations of heavily degraded crater segments is 35.5%, which shows that, similar to interpreting statistically derived modes, global averages alone do not indicate that a damage zone exists.

The global distribution of basin-radial and -concentric structures closely mimics that of the data sets from which these subsamples are extracted. Areas of higher fault or crater spatial densities are also found to return more basin-radial and -concentric segments and areas that are consistent with a wide range of basin-radial and -concentric also return more data. Furthermore, the geographic regions encompassing the poles display such a wide range that nearly all structures there are consistent with radial and concentric orientations, such that these regions were excluded from the analysis, and only the longest structures were manually selected and displayed for completeness. Similarly, all structural segments in the geographic regions covering the basin and the antipode were excluded from the analysis, as these regions were not subdivided further to allow for meaningful extraction of basin-radial and -concentric orientations. The antipodal region also contains chaotic terrain, which may further complicate interpretations.

We quantify the density of structural segments for each of our maps in Figure 4 as histograms displaying latitudinal and longitudinal counts of datapoints that we equally spaced in 1 km intervals along the structures. For comparison, we also produced the same histograms for the data neither considered radial nor concentric to the Caloris basin, representing a null model (Figure S4 in Supporting Information S1). The average of observations in the histograms is represented by a black line, thus showing where observations are above and below average. In

each histogram, the locations of the Caloris basin and its antipode are marked for reference. Since the geographic regions of the basin and antipode were excluded from the analysis, the number of observations in the histograms for these locations should not be interpreted as lulls in the total number of structures. For the concentric structures (Figure 4a), longitudinal distributions are generally near the average but show a symmetric increase halfway between the basin and the antipode to both the west and east. There is a slight increase in longitudinal concentric structure density just outside the eastern part of the basin. Latitudinally, the density of concentric structures increases north of the basin, which is likely due to a higher density of faults mapped in the smooth plains and presumably not indicative of a damage zone. The trends seen in the concentric data (Figure 4a) are not seen in the null model (Figure S4 in Supporting Information S1).

Figure 4b shows that there are more observations of radial structures than there are concentric ones, and their longitudinal and latitudinal distributions show a more distinct pattern. Longitudinally, the basin-radial structures are well above the spatial density average, directly adjacent to the Caloris basin. This pattern is mimicked at the antipode. The number of structures drops continuously and symmetrically away from the regions directly adjacent to the basin and antipode to below average. Interestingly, the longitudes showing below-average numbers of radial structures coincide with those of above average numbers of the concentric structures described in Figure 4a. Latitudinally, basin-radial structures are also well above average, directly adjacent to the Caloris basin with a similar trend holding true for the antipode. The trends seen in the radial data (Figure 4b) are not seen in the null model (Figure S4 in Supporting Information S1).

We calculated the proportions of basin-radial and -concentric structures with respect to the total number of structures for each geographic region for faults, as well as both straight crater-rim segments. We compared the percentages of fault segments and moderately degraded to fresh straight crater-rim segments individually to the percentage of the heavily degraded straight crater-rim segments in each geographic region. Geographic regions showing higher percentages in the data sets of fault segments and moderately degraded to fresh straight crater-rim segments were identified. In addition, we calculated the proportion of the range of basin-radial and -concentric orientations with respect to the total range of possible orientations to identify those geographic regions where the percentage of basin-radial and -concentric faults and moderately degraded to fresh straight crater-rim segments exceeded the percentage of the sampled range. We combined these two comparisons in bivariate maps (Figure 5). These maps show the geographic regions where structures are unlikely to be extracted because they were coincidentally consistent with basin-radial and -concentric orientations and thus allow us to interpret the extent of the Caloris basin damage zone.

Both bivariate maps contain many geographic regions where basin-radial and -concentric orientations exceed the percentages of the control distribution and the sampled range and thus, per our metrics, contain a signature of the Caloris basin damage zone. In both maps, the signature appears to be globally and unevenly distributed. For fault segments (Figure 5a), a strong signature is found to the southwest of the Caloris basin, which is consistent with the presence of strongly radial oriented faults (Figure 2; Figure S1 in Supporting Information S1). There is another strong signature to the north–northwest of the Caloris antipode. For moderately degraded to fresh straight crater-rim segments (Figure 5b), the strongest signatures are found in geographic regions to the west and southeast of the Caloris basin. Although the percentages for the geographic regions directly to the north of Caloris cannot be specified, it is of note that geographic regions to the northeast of the basin do not show a damage zone signature on either map (Figure 5).

4. Discussion

Our study investigated the extent to which the spatial density of faults and straight crater-rim segments reflect a damage zone caused by the impact forming the Caloris basin. We analyzed structural orientations and structure densities around the Caloris basin and assessed where the data are most likely to indicate a signature of this damage zone. The information we extracted can be put in the context of the predictions for impact damage zones pertaining to (a) fracture orientation, (b) density, and (c) spatial distribution of the damage zone.

4.1. Radial and Concentric Orientations Around the Caloris Basin

Impacts are known to form several types of fractures within the target rock surrounding the crater, including with radial and concentric orientations around the impact site and horizontal spall fractures (Ahrens & Rubin, 1993). Of those, radial and concentric fractures of vertical, inclined, or horizontal dips would all be suited for forming

straight rims of polygonal impact craters. Any inclined fractures, including the conical fractures that are found to be a subtype of the radial structures in a damage zone (Ahrens & Rubin, 1993), but especially horizontal spall fractures, would be well suited for reactivation as thrust faults by global contraction.

We assessed the orientations of straight rim segments of polygonal impact craters and faults across Mercury and extracted those segments with a matching orientation for further investigation. Specifically, we statistically determined the modes in the azimuth distributions of the structures as we binned them into geographic regions with a radial and concentric configuration around the Caloris basin geometric mean center. We found that more than one third of all structural segments in those data sets that could host fractures of a damage zone showed geometries consistent with an impact damage zone. Statistically, many of the strongest modes in our azimuth distributions were aligned with basin-radial and -concentric orientations (Figures 2 and 3a).

We also performed the same analysis on heavily degraded straight crater-rim segments. The craters from which these structural data were derived are superposed by geological units of the Caloris basin, such that they must be stratigraphically older and thus should not have recorded a damage zone from the Caloris basin-forming impact. This analysis revealed that not only global averages of the proportion of extracted basin-radial and -concentric orientations in this population closely match those of the faults and moderately degraded to fresh straight impact crater rims, but also that their statistically significant modes in the azimuth distributions are equally frequently aligned with the geometry of the damage zone when considering the entire planet. This finding shows that, if radial and concentric fractures formed around the center of Caloris, their proportion and statistical modes cannot be distinguished from a global average. But local comparisons, for example, specifically interpreting orientations for each geographic region, particularly near the basin, may still yield more detailed results. We also note that fractures in a damage zone, especially if rejuvenating preexisting structure, may not entirely form in perfectly radial and concentric orientations and thus there is uncertainty of the impact introducing radial and concentric orientations perfectly around the present-day geometric mean center of the Caloris basin. Moreover, multiple, superposed tectonic processes may share regions in which the resulting structural trends are identical. Individually and together, these reasons may serve to render basin-radial and -concentric orientations belonging to the damage zone difficult to distinguish from the multitude of superposed global tectonic patterns on Mercury.

Furthermore, we note that our analysis is unable to distinguish between radial and concentric orientations formed directly as impact damage or by post-impact basin-tectonic processes, including loading from basin infill (e.g., Kennedy et al., 2008). Both deformational processes—especially if loading were distributed axisymmetrically—would form structures of basin-radial and -concentric orientations and introduce the same structure type (i.e., joints and/or normal faults) in the area surrounding the basin (Kennedy et al., 2008), providing planes of weaknesses for fault reactivation or be utilized as rims of polygonal impact craters. Because such processes are related to the same basin-forming event, even asymmetries introduced during the impact process, such as from an oblique impact, would likely be similarly reflected in the geological record of both processes. However, post-impact basin-tectonic deformation would be found only in the areas adjacent to the basin and because of the identical geometry of the processes would very likely reactivate structures of the previously formed damage zone. Radial and concentric structures some distance from the basin are unlikely to have been caused by post-impact basin-tectonic processes.

4.2. Variations in Fracture Density Around the Caloris Basin

Impact damage generally shows a decrease in fracture density with distance from the impact site (Ahrens & Rubin, 1993), but field observations from simple and complex craters on Earth suggest that fracturing should be noticeably increased compared to the background fracture density within about 1–2 crater radii. We investigated the density of basin-radial and -concentric structures for faults and moderately degraded to fresh straight crater rims globally across Mercury by computing longitudinal and latitudinal frequencies showing where the spatial concentration of structures is greater than the background value (Figure 4). We show that especially for basin-radial structures, such frequencies are well above average directly adjacent to the Caloris basin in both latitudinal and longitudinal histograms. A similar pattern was found to exist for the antipode. The number of longitudinal observations is also found to drop continuously and symmetrically away from the basin and antipode to below average. Although we did not statistically evaluate for anti-correlation of the structure frequency data, we visually detected a longitudinal pattern that shows below-average numbers of radial structures coinciding with those of above average numbers of the concentric structures. Although the latitudinal and longitudinal frequencies

of structures do not capture the exact locations of the Caloris damage zone, they do show a pattern that is consistent with a high structure density directly adjacent to the basin and a decrease of structures away from it.

The cause of the below-average longitudinal radial structure counts coinciding with above-average counts of concentric structures is not directly apparent from our analysis, but if it is indeed related to an impact damage zone, it may imply basin-radial and -concentric fracturing to be preferentially occurring at different distances from the impact site. Xia and Ahrens (2001) noted that the furthest propagating fractures were radial in their experiments, which they attributed to extensional hoop stresses. It thus may be possible that, on the scales of large impact basins and the size of their parent bodies, this would result in seismic wave interference not only at the basin antipode but also promoting the formation of fractures of different orientations across a large part of the planetary body on which the large impact occurred.

In fact, the increase in observations toward the longitudes and latitudes of the antipode may indicate that seismic interference from the impact has indeed contributed to fracturing in these regions, affecting the structural fabric of the area. While the chaotic terrain near the Caloris basin antipode was not found to be caused by the basin-forming impact itself but by later modification utilizing a preexisting structural fabric (Rodríguez et al., 2020), seismic effects of the impact were likely global, and seismic waves may have interfered not only at the antipode but elsewhere, as well. Such seismic interference may also not have been following perfectly basin-radial and -concentric patterns, such that lithospheric fracturing and rejuvenation of preexisting structure, although increased toward the longitudinal and latitudinal regions of the basin and antipode, may have affected a wider range of orientations of structures than predicted by impact damage models from experiments and by insights from terrestrial craters. Future studies searching for large basin impact damage at other basins, perhaps on the Moon or Mars, or impact basin damage modeling studies should investigate the effects, geometry, and distribution of impact damage for such large basins.

4.3. Distribution of Damage Around the Caloris Basin

The symmetry of damage zones around impact craters is thought to indicate the direction of the impactor. Symmetric damage zones around the crater are consistent with impacts with high angles, whereas oblique impacts would produce asymmetric damage distribution around the crater, showing more damage in the downrange direction (Kenkmann et al., 2014). We assessed which of the geographic regions that we binned our observations into is likely to show a signature of the Caloris basin damage zone (Figure 5). The analysis revealed that there are signatures consistent with the predictions for impact damage zones in several geographic regions, but the majority of regions did not contain a signature. Strong signatures were located in those geographic regions that are directly adjacent to the west, southwest, and southeast of the Caloris basin. These same regions also showed the most obvious orientation pattern, where the strongest modes were aligned with basin-radial or -concentric orientations. We therefore interpret those regions as the location of the Caloris basin damage zone.

Although our method did not allow us to specify percentages for the geographic regions directly to the north of the Caloris basin, we noted that the geographic regions to the northeast of the Caloris basin do not show a damage zone signature (Figure 5). We tentatively interpret the distribution of the signature of damage as asymmetric. This asymmetry implies that the Caloris basin formed from an oblique impact, with the impactor coming from the northeast. The asymmetry and thus the direction of impact are further supported by previous geologic observations, as the overall shape of the Caloris basin was inferred to be elliptical with its long axis oriented in the ENE–WSW direction (Fassett et al., 2009) and the distributions of impact-ejecta-related knobs were found to be most numerous and most distal southwest of the Caloris basin (Wright et al., 2020).

The regions that were excluded from our analysis were those that showed a very wide range of orientations consistent with basin-radial and -concentric geometries. Further division of these geographic regions into smaller regions or automation of the methodology, perhaps with machine-learning algorithms, would improve the precision of our method, which could expand the analysis to the uninterrogated regions (Figure 5). It should also be noted that we searched if a pattern existed using a hypothesized geometry in a predefined geometric configuration, but with data containing several different preferred orientations originating from multiple patterns. Automated pattern recognition could investigate systematic orientations and changes thereof across Mercury, accounting for the overlap of multiple patterns including those from earlier, large basins on the planet.

Lastly, it is important to consider that damage zone formation of this scale is poorly understood and that for this reason there are no specific predictions for the presence of damage, expected orientations of damage zone structures in, and the effects of oblique large impacts on the antipodal hemisphere. An oblique impact trajectory could shift the far-field focusing and convergence away from the geometric antipode. Furthermore, as noted above, the nature of seismic wave interference may also form damage zones with structures of orientations not consistent with radial and concentric orientations to the impact site and it is unknown in what geometries and to what areal extent these interferences would contribute toward forming a damage zone. These gaps in knowledge in our scientific field should be addressed in future numerical simulations. The geological record of the Caloris basin-forming impact and the wealth of globally-distributed tectonic structures on Mercury are ideal for addressing the many unknowns associated with impact damage formation during very large meteoritic impacts.

5. Conclusions

Mercury is replete with tectonic landforms (Byrne et al., 2018) (Figure 1). Previously, a global compilation of faults and analysis of their orientations determined that Mercury patterns reflect multiple overlapping tectonic processes (Klimczak, Crane, & Byrne, 2025). One of the processes hypothesized to have contributed to the pattern was the reactivation of damage zone fractures from the Caloris basin-forming impact by global contraction, which explained many of the preferred orientations in the fault data. We further tested this hypothesis by including straight rim segments of polygonal impact craters that are also commonly found across Mercury (Yazıcı et al., 2024). The crater data are especially useful because they contain stratigraphic information that allowed us to separate them into two populations, one in which crater degradation characteristics are consistent with superposition by Caloris basin geological units, and the other that show crater degradation characteristics consistent with them superposing Caloris basin geological units. Because polygonal impact craters form by utilizing pre-existing structures, the more heavily degraded craters cannot have utilized impact damage associated with the Caloris basin, whereas the moderately degraded to fresh craters would have utilized fractures of such impact damage if it were preserved in Mercury's lithosphere.

Our analysis reveals that statistical modes of Caloris basin-radial and -concentric orientations that previously led to the hypothesis of a widespread damage zone being reflected in the fault data are similarly prominent in the straight rim segments of impact craters, including the population of heavily degraded ones. This shows that the presence of a damage zone cannot be confirmed solely by interpretation of statistical results of the orientations of tectonic structures. We isolated structural segments from faults and moderately degraded them to fresh craters for additional analysis. We do find a signature of a damage zone in the spatial density of structures and identify geographic regions where basin-radial and -concentric orientations are likely to exceed the number of observations reflecting coincidentally sampled orientations due to the natural variation of the data (Figure 5). We tentatively interpret the Caloris basin damage zone to be asymmetric, showing higher intensities and greater preservation to the east, southwest, and southeast, with no definitive signature to the northeast. Together with observations of a slightly elliptical basin shape (Fassett et al., 2009) and distribution of ejecta (Wright et al., 2020), the damage zone asymmetry is consistent with an oblique impact and the impactor arriving from the northeast.

The straight rims of impact craters on Mercury show multiple preferred orientations, and strong modes in their orientation distributions indicate that there are structural patterns preserved in the data. Future studies should complement their analysis with the straight crater rim data set (Yazıcı et al., 2024), and crater rim orientations should be compared in more detail among the different subpopulations and to the fault data to gain better insight into the tectonic patterns and underlying processes on Mercury. Similarly, the method used in this paper can be applied to other planetary bodies hosting global populations of faults and polygonal impact craters. Mars and the Moon, in particular, are heavily cratered with a noticeable portion of polygonal impact craters and a wealth of fault-related landforms that could be used to investigate damage zones around their largest impact structures.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

All data used in this research are publicly available. The fault data is published in Klimczak, Byrne, and Crane (2025). The crater data is available in Yazıcı et al. (2021). The data for the Caloris basin data is published in Byrne and Klimczak (2025). We make the map data from our geospatial analysis in this study is publicly available in Klimczak et al. (2026).

Acknowledgments

The authors thank Debra Buczkowski for the editorial handling of the manuscript. We acknowledge Teemu Öhman and an anonymous reviewer for their constructive feedback on this paper.

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